

Polish closure: the analytic cross-recombination pin and endpoint hardening (gate STEP-5B, post-closure quality items)

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

Operator verdict #14 flipped gate STEP-5B to CLOSED-CONDITIONAL and left two polish items: the analytic pin of the cross-recombination constant c_{cross} and the hardening of the thin endpoint margin ($\times 2.1$ at $I = 2 \times 10^{-3}$). This note closes both. Scope: matched second-cumulant B5 bookkeeping, as everywhere in the chain.

2. The analytic cross-recombination pin

Theorem 1 (Curvature pins c_{cross} — depth-free). *Let P' refine the readings of an admissible P into sub-resolution clusters. Cross-cluster recombination — two distinct ordered pairs $(\hat{u}_i, \hat{v}_j) \neq (\hat{u}_{i'}, \hat{v}_{j'})$ from cap pair (r, r') landing on the SAME transfer — requires the exact vector identity $\hat{u}_i - \hat{u}_{i'} = \hat{v}_{j'} - \hat{v}_j$. If a nontrivial family of such identities holds, the two cap point-sets share a difference set, i.e. one is a translate of the other within the sphere; since $S \cap (S + w)$ is a circle, BOTH cap sets lie on one common circle. Therefore: (i) for non-co-circular clusters the exact recombination count is the trivial ordering/sign degeneracy (≤ 4 ordered) and the cross ℓ^2 is EXACTLY preserved under splitting; (ii) co-circular clusters are precisely the configurations covered by the universal single-circle theorem ($K = 14$, sharp, any amplitudes). Consequently the energy-faithfulness constant of the AddC/AddD chain is depth-free:*

$$\Delta E \leq \underbrace{6I^2}_{\text{within-cap saturation}} + \underbrace{14I^2}_{\text{co-circular, absorbed}} = 20I^2,$$

with no dependence on cluster depth k or alignment engineering.

Machine audit: adversarial aligned-AP caps (5+5 points, two distinct cap planes) show max EXACT fiber multiplicity 2; the co-circular control (one circle) shows multiplicity 6 with measured $K = 12.20 < 14$. The would-be “alignment pumping” of cross energy (which grows linearly in depth k for ROUNDED coincidences) is an artifact of finite tolerance: at exact arithmetic, curvature splits the fibers at $O(\delta^2)$, and the free-energy functional matches transfers exactly.

3. Endpoint hardening

Proposition 1 (Criterion band and the J -refinement). *(i) The resolution criterion $\theta_{\min} = c_\theta/(q_0\xi)$ with $c_\theta \in [1, \pi]$ has its closure-relevant CONSERVATIVE end at $c_\theta = 1$ (the current choice): larger c_θ shrinks the admissible class and thickens every margin, so the quoted margins are FLOORS*

with respect to criterion ambiguity. (ii) Amended-class patterns have pairwise separations $\geq \theta_{\min}$, hence all transfers obey $|t| \geq 2q_0 \sin(\theta_{\min}/2)$, and the envelope weight improves from $J_{\max} = 0.273$ to $J_{\text{eff}} = J(2q_0 \sin(\theta_{\min}/2))$: 0.256 at the production anchor, 0.226 at the endpoint. The closure margins become

$$\times 59.4 \quad (I = 4 \times 10^{-4}), \quad \times 2.6 \quad (I = 2 \times 10^{-3}, \text{ floor}),$$

with the π -phase end of the band at $\times 290.9$ and $\times 12.7$ respectively.

4. Consequence for the closure record

Both verdict-#14 polish items are closed: the AddD closure margins are now $\times 59.4 / \times 2.6$ (floor) at the two band-edge intensities, the constant chain is depth-free and analytic throughout, and no machine constant in the closure path remains unpinned except the PROVISIONAL 20/9 incidence route (which the closure does not use).

5. Numerical verification

Script [codes/vacuum/beyond_layer_gershgorin_bound.py](#) (v1.14.0; 192/192 self-test asserts PASS; artefact [claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/result.json](#)).

Quantity	Value	Check
adversarial exact multiplicity	2	trivial degeneracies only
co-circular control	multiplicity 6; $K = 12.20 < 14$	absorbed by single-circle theorem
c_{total}	≤ 20 , depth-free	analytic pin
J_{eff} (anchor / endpoint)	0.256 / 0.226	vs $J_{\max} = 0.273$
hardened margins	$\times 59.4 / \times 2.6$ (floors)	band tops $\times 290.9 / \times 12.7$

Quantitative sanity check (mandatory): the endpoint improvement ratio $2.6/2.1 = 1.24$ is close to $J_{\max}/J_{\text{eff}} = 0.273/0.226 = 1.21$ (small residual from rounding in the n_{pack} floor), dimensionally a pure ratio as required; the co-circular control's $K = 12.20$ obeys the sharp bound $14 - 18/n = 12.20$ at $n = 10$ EXACTLY, confirming the absorption argument with zero slack.

6. Devil's-advocate (three objections; CLAUDE.md 6.3)

(α) “Near-coincident fibers (split by $O(\delta^2)$) still interact through the smooth $J(|t|)$ envelope, even if exact recombination vanishes.” DISMISSED with the chain's own structure: the ℓ^2 energy $\sum_t w_t^2$ counts EXACT fibers (Parseval is exact); the envelope J weights each fiber separately and J is bounded by $J_{\text{eff}} < J_{\max}$ on every realized transfer — near-coincidence affects neither count nor weight beyond what is already bounded.

(β) “The criterion-band argument inverts the burden: maybe the TRUE resolution scale is below $1/(q_0\xi)$, enlarging the class.” VALID-with-mitigation: $1/(q_0\xi)$ is the standard coherence estimate (beat phase ~ 1 over one correlation length); a criterion BELOW it would assert that two modes whose beat cannot complete even one radian within a correlation volume are independently resolvable, which contradicts the indistinguishability lemma's energy-faithfulness (restructuring at that scale shifts F by margin/ $\times 33$ or less — the physics cannot tell them apart). The honest floor stands at $c_\theta = 1$.

(γ) “The endpoint floor $\times 2.6$ still depends on the $\times 4$ -free packing count $16/\theta_{\min}^2$.” DISMISSED: $16/\theta^2$ IS the conservative cap-packing bound ($2/(1-\cos(\theta/2)) \approx 16/\theta^2$ to 0.7% at these angles, machine-checked in AddC); no looser constant enters.

7. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND / AddE (gate STEP-5B)
Precise statement:	(a) c_cross ANALYTIC PIN (depth-free): exact recombination forces co-circularity; curvature splits all other alignments; c_total $\leq 20 I^2$. (b) ENDPOINT HARDENING: criterion band has its conservative end at the current theta_min; J-refinement ($J_{\text{eff}} = 0.256/0.226$) lifts the closure margins to $\times 59.4$ / $\times 2.6$ (floors), band tops $\times 290.9$ / $\times 12.7$.
Scope:	Matched second-cumulant B5 bookkeeping; amended class of verdict #14.
Dependencies:	AddC/AddD (lemma + closure record); universal single-circle theorem (main chain v1.8); J machinery (main chain).
Evidence grade:	ANALYTIC (both pins) + EXECUTED (192/192)
Reproduction command:	python codes/vacuum/beyond_layer_gershgorin_bound.py
Expected output:	claims 192/192 PASS; result.json under claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/
Falsification gate:	A non-co-circular cluster pair with exact fiber multiplicity > 4 (refutes the curvature pin); an amended-class transfer with $J(t) > J_{\text{eff}}$ (refutes the refinement).
Tier before / after:	AddE enters at T4; B5 remains T5-CANDIDATE (verdict #14); gate remains CLOSED-CONDITIONAL.
No-overclaim statement:	The polish strengthens the closed gate’s margins; it does NOT change the conditional character of the closure (amended class), nor any tier. The 20/9 incidence route stays provisional and unused. Unrestricted class remains open (DR-2, research-grade).
Next required action:	Mainline: Reading-H T6 discussion (gate consequence). Optional: DR-2; full T5 assignment for B5 at operator review.